

8 (a) (i) $(F =) -kx$

(ii) By Newton's second law of motion,

$$F_{\text{net}} = ma$$

$$-kx = ma$$

$$a = -\frac{k}{m}x$$

Since both k and m are constants ($\frac{k}{m}$ is a constant), a and x satisfy the defining equation of simple harmonic motion i.e., $a = -\omega^2 x$ (where ω is constant), and the block oscillates with simple harmonic motion.

- (iii) Comparing the equation obtained with $a = -\omega^2 x$,

$$\omega^2 = \frac{k}{m}$$

$$\left(\frac{2\pi}{T}\right)^2 = \frac{k}{m}$$

$$T = 2\pi\sqrt{\frac{m}{k}} \quad (\text{shown})$$

- (b) (i) 1. Let the speed right after collision be v , by principle of conservation of linear momentum,

$$m_A u_A + m_B u_B = m_A v_A + m_B v_B$$

$$m_B u_B = (m_A + m_B) v \quad (\because u_A = 0)$$

$$v = \left(\frac{m_B}{m_A + m_B}\right) u_B$$

$$= \left[\frac{18 \times 10^{-3}}{(35 + 18) \times 10^{-3}}\right] (5.3)$$

$$= 1.80 \text{ m s}^{-1}$$

By principle of conservation of energy,
loss in kinetic energy = gain in elastic potential energy

$$\frac{1}{2} (m_A + m_B) v^2 = \frac{1}{2} k x_{\max}^2$$

$$x_{\max}^2 = \frac{(m_A + m_B) v^2}{k}$$

$$x_{\max} = \sqrt{\frac{m_A + m_B}{k}} v$$

$$= \sqrt{\frac{(35 + 18) \times 10^{-3}}{25}} (1.80)$$

$$= 0.0829 \text{ m}$$

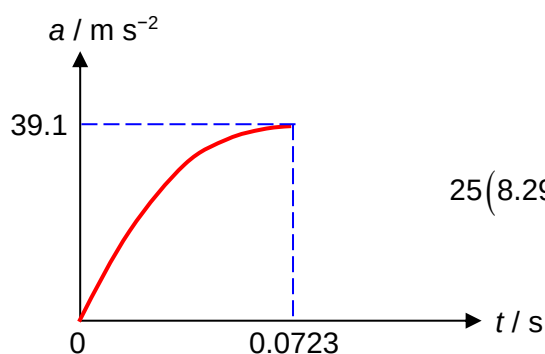
2. $t = \frac{1}{4} T$

$$= \frac{\pi}{2} \sqrt{\frac{m_A + m_B}{k}}$$

$$= \frac{\pi}{2} \sqrt{\frac{(35 + 18) \times 10^{-3}}{25}}$$

$$= 0.0723 \text{ s}$$

(ii)



$$F = ma$$

$$kx_{\max} = ma_{\max}$$

$$25(8.29 \times 10^{-2}) = [(35 + 18) \times 10^{-3}] a_{\max}$$

$$a_{\max} = 39.1 \text{ m s}^{-2}$$

- (c) (i) Considering horizontal motion of block B, by Newton's second law of motion,

$$F_{net} = ma$$

$$f_0 = m_B a_0$$

$$a_0 = \frac{f_0}{m_B}$$

$$= \frac{9.5 \times 10^{-2}}{18 \times 10^{-3}}$$

$$= 5.2778 \text{ m s}^{-2}$$

Considering horizontal motion of both blocks, by Newton's second law of motion,

$$F_{net} = ma$$

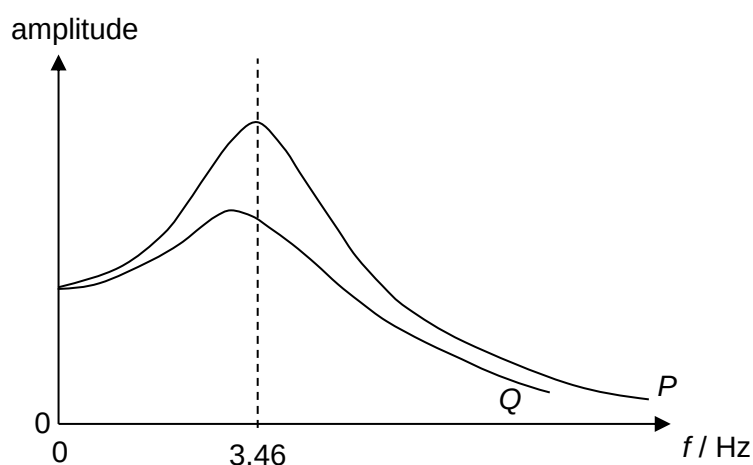
$$-kx = (m_A + m_B)a$$

$$a = -\left(\frac{k}{m_A + m_B}\right)x$$

$$\begin{aligned} x_0 &= \left(\frac{m_A + m_B}{k}\right)a_0 \\ &= \frac{[(35 + 18) \times 10^{-3}](5.2778)}{25} \\ &= 0.0112 \text{ m} \end{aligned}$$

- (ii) 1. resonance occurs when driving frequency (or f) equals (or close to) natural frequency of oscillations of the blocks causing large amplitude of oscillations

2.
3.



$$\begin{aligned} T &= 2\pi\sqrt{\frac{m}{k}} \\ f &= \frac{1}{2\pi}\sqrt{\frac{k}{m}} \\ &= \frac{1}{2\pi}\sqrt{\frac{25}{(35 + 18) \times 10^{-3}}} \\ &= 3.46 \text{ Hz} \end{aligned}$$

For Q:

Correct shape and peak of Q lower than P and the left of P

9 (a) Charge distribution: Some of the alpha particles were observed to be scattered at